

Energy Resources

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Energy:

- Energy is the ability to do work.
- Energy consumption of a nation is usually considered as an index of its development as almost all activity directly or indirectly dependent on energy.
- When energy is converted in to electricity it is called power.

Growing energy demand:

- Cooking , Heating , lighting, mining, agriculture
Industrial activities all need energy.
- Transporting people and goods.
- Manufacturing consumer goods and capital equipment.
- With the growing population the demand of energy is also increases.

Energy Resources

The energy resources can be classified in many ways:

A) Commercial fuels-These include coal, lignite, petroleum products, natural gas and electricity.

Non-commercial fuels- These include fuelwood, cow-dung, agricultural wastes, etc.

B) Primary Energy Resources-These are those which are mined or otherwise obtained from nature.

Secondary Energy Resources: They are those which do not occur in nature, they are derived from primary energy resources. These include petrol, diesel.

C) Conventional Energy Resources: These include fossil fuels(coal, petroleum and natural gas), water and nuclear energy.

Non-conventional Energy Resources: These include solar, wind, geothermal, ocean , biomass and hydrogen energy.

D) Non-Renewable Energy Resources: Those resources which are exhaustible and cannot be replaced once they are used. These are available in limited amount and develop over a long period. These include fossil fuels and nuclear power.

Renewable Energy Resources: Those resources which are inexhaustible and can be used to produce energy again and again. These include solar, wind, water, geothermal, ocean and biomass energy.

Renewable Energy Resources to be studied:

- Solar Energy
- Wind Energy
- Hydro Energy
- Biomass Energy
- Geothermal Energy
- Ocean Energy Systems
 1. Tidal Energy
 2. Wave Energy
 3. Ocean Thermal Energy

Non-Renewable Energy Resources

- Coal
- Petroleum
- Natural Gas

Solar Energy:

- Solar energy is the radiant light and heat from the sun.
- Solar energy is the energy received by the earth from the sun.
- Solar power is the conversion of sunlight into electricity.

Solar energy is produced by using range of solar energy harnessed technologies.

1) Photovoltaic Plants (PV) : Solar cells

2) Concentrating Solar Thermal Plant (CSP): Solar thermal technology, solar architecture ; solar power plant

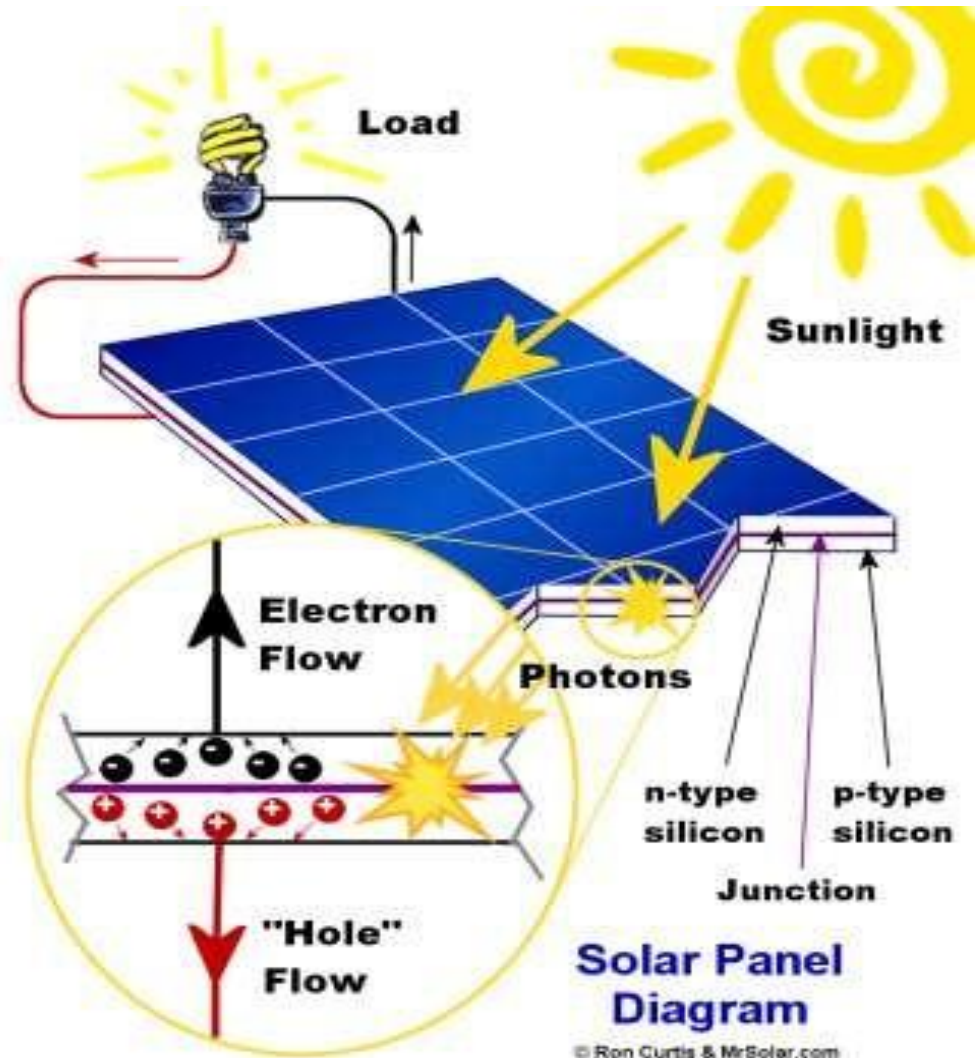
- **Solar heating technology : solar cooker, solar heater**

1) Photovoltaic Plants (PV) : Solar cells

- Solar cells are used to convert solar heat into electricity are called photovoltaic cells.
- Solar cells are made up semiconducting materials like silicon , germanium.
- When solar radiations strike them, electrons are knocked out loose from the molecule , and therefore generation of positive and negative charges takes place. The charge separation is developed across the junction between them.
- P-type silicon conducts positive charge while n-type silicon conducts negative charge.
- When the two surfaces are joined by a connector like a wire , a potential difference is produced between both the surfaces of the solar cell which causes flow of electrons and produce electricity.

1) Photovoltaic Plants (PV) : Solar cells

- The potential difference produce by a single PV cell of 4cm² size is about 0.4-0.5 volt which produces a current of 60 milli amperes.
- A group of solar cells joined together in a definite pattern form a solar panel .
- **Solar panel** can harness a large amount of solar heat and can produce electricity enough to run **street light , irrigation water pump** etc.



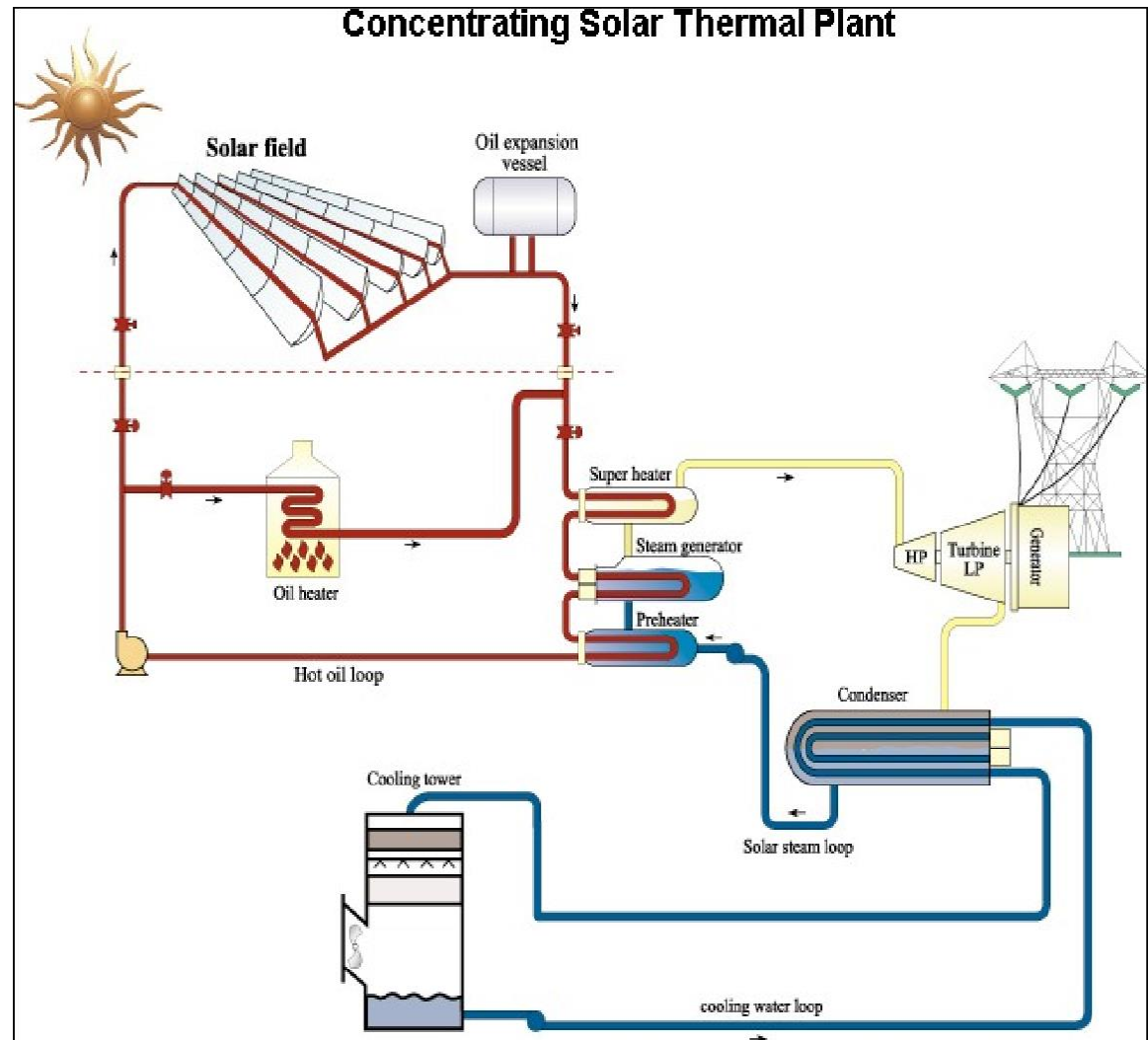
- Solar cells turn sunlight into electricity in an environmentally clean manner(sustainable energy resource).
- Initial cost is high as silicon wafers are used to make solar cells and silver used to connect no. of cells to form solar panel.
- They can be installed in remote areas, in forests and deserts where installation of electric cables are cost –prohibitive.
- Electricity in Ladakh in the Himalayas is generated using solar panels(Large number of solar cells , put together form a solar panel).
- This electricity produceds used for various purposes household and industrially.
- There are several companies in India that offer solar home –lightening systems.

2. Concentrating Solar Thermal Plant (CSP)

- CSP offers advantages for **central and large-scale** applications. CSP power plants are the most **cost-efficient**.
- **way to generate and to store dispatchable CO₂ free electricity.**

It contains:

- **Collector Field**
- **Turbine Generator**
- **Cooling Tower**
- **Transformer**



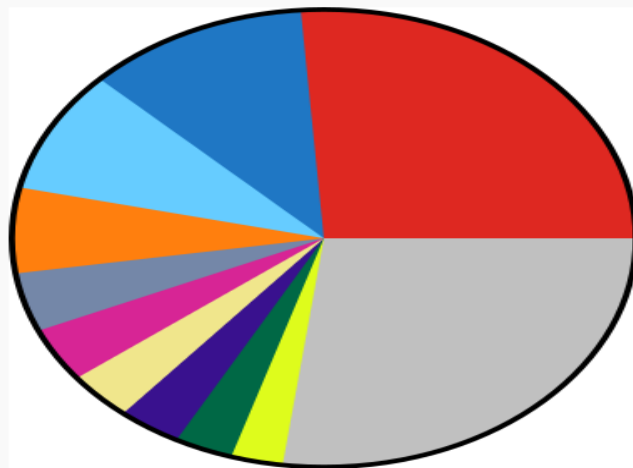
Solar Thermal Technology

This uses heat gained directly from sunlight. The best known use of this technology is in :

- Solar water heating**
- Solar heating of buildings**
- Solar dryer for food grains and other agricultural products**
- Solar distillation (for water purification)**
- Solar cooker (For cooking)**
- Solar powered vehicles**
- Solar thermal power generation**

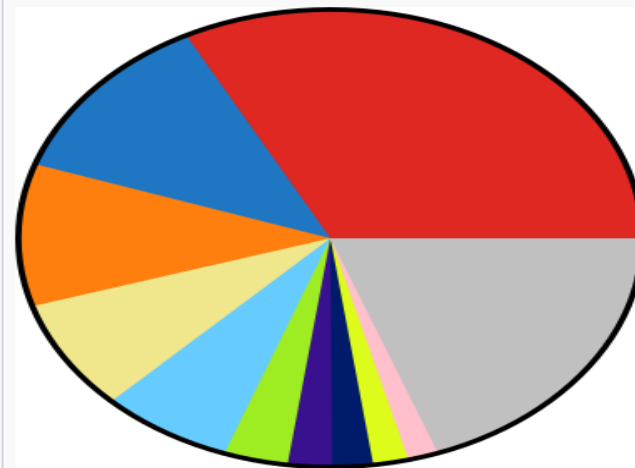
Solar Energy – Current Scenario

Top 10 countries by added solar PV capacity in 2019^[13]



China	30,100 MW (26.2%)
United States	13,300 MW (11.6%)
India	9,900 MW (8.6%)
Japan	7,000 MW (6.1%)
Vietnam	4,800 MW (4.2%)
Spain	4,400 MW (3.8%)
Germany	3,900 MW (3.4%)
Australia	3,700 MW (3.2%)
Ukraine	3,500 MW (3.0%)
South Korea	3,100 MW (2.7%)
All others	31,200 MW (27.2%)

Top 10 countries by cumulative solar PV capacity in 2019^[14]



China	204,700 MW (32.6%)
United States	75,900 MW (12.1%)
Japan	63,000 MW (10.0%)
Germany	49,200 MW (7.8%)
India	42,800 MW (6.8%)
Italy	20,800 MW (3.3%)
Australia	14,600 MW (2.3%)
United Kingdom	13,300 MW (2.1%)
South Korea	11,200 MW (1.8%)
France	9,900 MW (1.6%)
All others	121,600 MW (19.4%)

Solar Energy – Current Scenario total expansion in Asia

- **India achieved 5th global position in solar power deployment by surpassing Italy.** Solar power capacity has increased by more than 11 times in the last five years from 2.6 GW in March, 2014 to 35 GW in July, 2019.
- Total solar capacity has now reached about the same level as wind capacity thanks largely to expansion in Asia (78 GW) in 2020.
- Major capacity increases in China (49 GW)
- Viet Nam (11 GW).
- Japan also added over 5 GW.
- India expanded solar capacity by more than 4 GW.
- Republic of Korea 4 GW.
- The United States of America added 15 GW

Wind Energy :

- Fast moving air is called wind . Winds are produced when differential heating of earth surface creates air masses with different heat densities.
- Differential heating of the earth's surface and atmosphere induces vertical and horizontal air currents that are affected by the earth's rotation and contours of the land and generates WIND.
- This kinetic energy possessed by air due to its velocity is called wind energy. This can be used to do work.
- This energy is harnessed in the form of Wind mills, which is used to generate electricity.
- To generate electricity , the rotatory motion of the windmill is used to turn the turbine of the electric generator.

Wind Farm

- A wind mill is used to turn the turbine consists of a structure similar to a large electric fan that is erected at some height on a rigid support.
- The output of the single windmill is quite small and cannot be used for commercial purposes.
- Therefore, a number of Windmills are erected over a large area which is known as **Wind Energy Farm**.



A picture of a Wind Energy Farm

Merits of Wind Energy

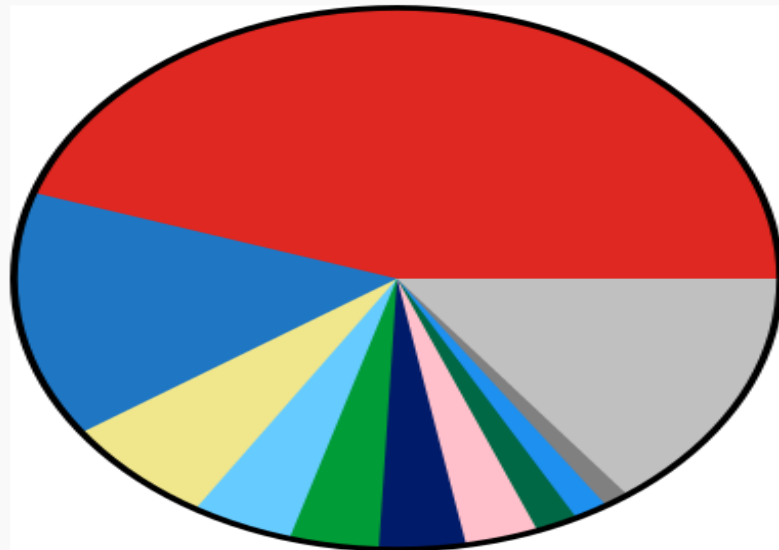
- It is non-polluting, environmental friendly and sustainable source of energy.
- The gestation period is low and the power generation starts immediately after commissioning of the windmill.
- Power generation is cheaper as there is no shortage of input i.e wind.

Limitations of Wind Energy

- Wind energy farms can be located only in vast open areas in favourable wind conditions. The minimum velocity for a wind mill to function is 15km/hour.
- The cost of construction of a wind energy farm is high.
- The continuous whirling and whistling is irritating.
- The location of the wind energy farms should not be on the routes of migratory birds as it will play havoc with these birds.
- There should be some backup facility to take care of the energy needs during a period when there is no wind.

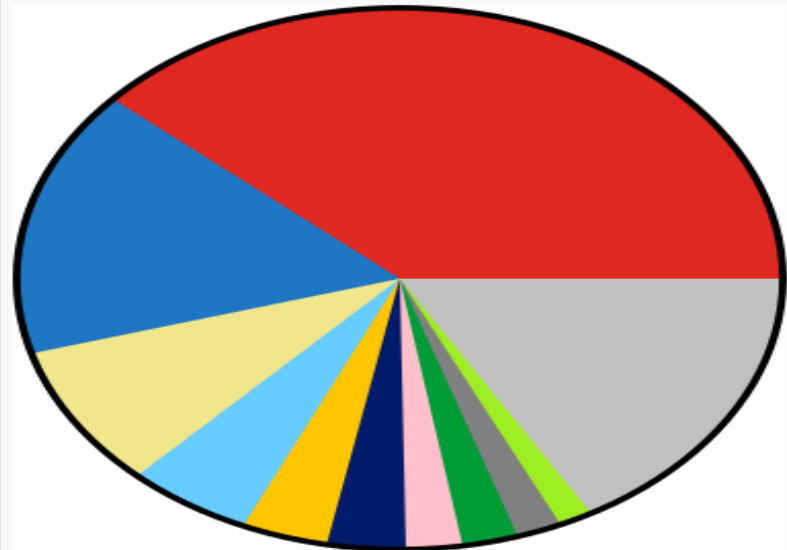
Wind Energy – Present Scenario

Top 10 countries by added wind capacity in 2018^{[13][14]}



China	23,000 MW (44.8%)
United States	7,588 MW (14.8%)
Germany	3,371 MW (6.6%)
India	2,191 MW (4.3%)
Brazil	1,939 MW (3.8%)
United Kingdom	1,901 MW (3.7%)
France	1,565 MW (3.0%)
Mexico	929 MW (1.8%)
Sweden	720 MW (1.4%)
Canada	566 MW (1.1%)
Rest of the world	7,546 MW (14.7%)

Top 10 countries by cumulative wind capacity in 2020^[15]



China	281,993 MW (38.5%)
United States	117,744 MW (16.1%)
Germany	62,184 MW (8.5%)
India	38,559 MW (5.3%)
Spain	27,089 MW (3.7%)
United Kingdom	24,665 MW (3.4%)
France	17,382 MW (2.4%)
Brazil	17,198 MW (2.3%)
Canada	13,577 MW (1.9%)
Italy	10,839 MW (1.5%)
Rest of the world	122,046 MW (16.6%)

Wind Energy – Present Scenario of India

- **Wind power generation** capacity in **India** has significantly increased in recent years.
- **As of 28 February 2021, the total installed wind power capacity was 38.789 GW, The fourth largest installed wind power capacity in the world.**
- **Wind power** capacity is mainly spread across the Southern, Western and Northern regions

Wind Energy – Present Scenario of India

- **Tamil Nadu** tops the list of states with the largest installed wind power generation capacity in the country .
Total wind capacity at the end of **2018 stood at 8,631 MW** while **its total installed electricity generation capacity stood at 30,447 Mw , with wind sector's share at 28.34 per cent.**
- Gujarat the second-largest state's total wind capacity at the end of 2018 stood at 6,044 Mw while its total installed electricity generation capacity stood at 31,382 MW at the end of 2018, with the wind sector's share at 19.25 per cent.
- Maharashtra the third-largest state's wind capacity at the end of 2018 stood at 4,789 Mw while its total installed power generation capacity stood at 43,779 Mw at the end of 2018.

Hydroelectricity

- Hydroelectricity is a form of electricity which is produced by water power. For eg: dams...
- the total electric power potential of India is 44594 MW or 13% of its total electricity utility .
- It is estimated to be 148700MW . Can meet the demand of around 85000MW.
- 80% of the hydel projects lie in Maharashtra, Punjab, Himachal Pradesh, Jammu and Kashmir and Uttaranchal.
- India 7th largest producer of hydropower.

Advantages:

- 1. Once a dam is constructed, electricity can be produced at a constant rate.**
- 2. The lake's water can be used for irrigation purposes.**
- 3. It is pollution free and the most versatile source of energy of all the known energy resources.**
- 4. The life expectancy of a hydropower plant is about 50 years or more.**

Disadvantages:

- 1. Dams are extremely expensive to build and must be built to a very high standard.**
- 2. Requirement of large areas of land means that the natural environment is destroyed.**

BIOMASS

Biomass is the common name for organic materials used as renewable energy sources such as; wood, crops, and waste.

Working of a Biogas Plant:

The working of Biogas plant involves cow dung, crop, residues, vegetable waste, poultry droppings and sewage are decomposed in the absence of oxygen (anaerobic decomposition) to give Biogas.

BIOMASS

- It is an excellent fuel as it contains up to 75% methane.
- It burns without smoke, leaves no residue and has high heating capacity.
- The slurry left behind is removed periodically and is used as excellent manure, rich in nitrogen, phosphorous and potassium.
- Biogas has a high calorific value (5000-5500 Kcal/Kg) and can be used for domestic purposes, operating small scale
- Industries, lighting and as fuel for the boilers, generators etc

First generation bio-fuels

- Bio-alcohols
- Biodiesel
- Green diesel
- Vegetable oil
- Bio-ethers
- Biogas
- Syn-gas
- Solid biofuels

Geothermal Power

- It is thermal energy generated and stored in the Earth as hot water springs.
- The Geothermal gradient, which is the difference in temperature between the core of the planet and its surface, drives a continuous conduction of thermal energy in the form of heat from the core to the surface. originates from the formation of the planet (20%) and from radioactive decay of minerals (80%).
- Geothermal power is considered to be sustainable because the heat extraction is small compared with the Earth's heat content.
- The oldest known spa is a stone pool on China's Lisan mountain built in the Qin dynasty in the 3rd century BC, at the same site where the Huaqing Chi palace was later built.

India has vast potential for geothermal power. North Western Himalayas and the western coasts are considered geothermal areas.

The geological Survey of India has already identified more than 350 hot water springs with average temperatures of 80°C to 100°C.

A 5 kW geothermal pilot power plant has been commissioned at Manikaran in Himachal Pradesh.

The Puga Valley in Ladakh region has the most promising geothermal field with a potential of 4.5MW power.

Merits of Geothermal Energy

- Geothermal power is cost effective, reliable, least polluting sustainable, and environmentally friendly.
- Geothermal power has the potential to help mitigate [global warming](#) if widely deployed in place of fossil fuels.
- As compared to solar energy and wind energy, the power generation level of geothermal energy is higher.

Limitations of Geothermal Energy

- Geothermal hot spots are scattered so locating these hot spots is really difficult. has historically been limited to areas near tectonic plate boundaries.
- Though, geothermal energy is inexhaustible, a single bore has a limited life span of about 10 years.
- Noise pollution is caused by drilling operations at geothermal sites.

Ocean Energy Systems

- Oceans are the large water bodies covering about 75% of the Earth`s total surface area. Apart from being large reservoirs of water, they are huge reservoirs of energy also.
- Oceans store energy which can be obtained for useful purposes in the following ways:
 1. **Tidal Energy**
 2. **Wave Energy**
 3. **Ocean Thermal Energy**

Tidal Power

Tidal power, also called tidal energy, is a form of hydropower that converts the energy of tides into useful forms of power - mainly electricity.

- Water level near the coasts rises up and falls twice a day. This movement of water level along the coasts is known as Tides.
- Tides are due to gravitational pull of the moon on waters in the ocean. This pull varies during the monthly cycle rotation of the moon around the earth.
- High tides occur on every new moon day and full moon day
- and can raise the level of water by few meters.
- Tidal energy is harnessed by constructing a dam across a narrow opening to the sea.
- A turbine fixed at the opening of the dam converts tidal energy to electricity.

Tidal Power

Advantages of Tidal Energy:

- It is an inexhaustible source of energy.
- Tidal energy is environment friendly energy and doesn't produce greenhouse gases.

Disadvantages of Tidal Energy:

- Cost of construction of tidal power plant is high.
- There are very few ideal locations for construction of plant and they too are localized to coastal regions only.

Wave Power

Wave power is the transport of energy by ocean surface waves, and the capture of that energy to do useful work.

Advantages of Wave Power

- Capable of high efficiency (60-80%) in ideal conditions.
- Renewable energy source obtained by wind via the Sun's heating of our atmosphere.

Disadvantages of Wave Power

- Improperly placed wave power plants can damage the marine ecosystem.
- Efficiency drops significantly in rough weather due to safety mechanisms.

Ocean Thermal Energy

- Large amount of solar energy is stored in the oceans and seas.
- On an average, 60 million square kilometre of tropical seas absorb solar radiation which is equivalent to the heat content of 245 billion barrels of oil.
- If this energy can be tapped, a large amount of energy will be available to the tropical countries and other countries as well.
- The process of harnessing the ocean thermal energy is popularly known as Ocean thermal energy conversion(OTEC).

Merits of OTEC :

- The electric power from OTEC is continuous, renewable and pollution free.
- It is one of the most clean power production technologies.
- These systems transfer nutrients from the unproductive deep waters to the warmer surface thereby enriching the fishing grounds.

• Limitations of OTEC:

- This process requires lot of capital investment.
- Due to small temperature difference between the surface water and the deep water, the conversion efficiency is low (3-4%).

OTEC potential in India :

- India has large potential of OTEC which could be of the order of 50,000 MW.
- Some of the best sites for this process plant are situated near the Lakshadweep, Andaman and Nicobar Islands.
- An OTEC plant is proposed off the coasts of Tamil Nadu.

Non Renewable Energy Sources

- **Crude Oil :** These petroleum products include gasoline, distillates such as diesel fuel and heating oil, jet fuel, petrochemical feedstocks, waxes, lubricating oils, and asphalt.
- **Natural Gas:** Natural gas consists mainly of methane, it can be used as a fuel or to make materials and chemicals.
- **Coal:** It's a sedimentary rock with a high amount of carbon and hydrocarbons.
- **Uranium (Nuclear Energy):** Uranium is the fuel most widely used by nuclear plants for nuclear fission. Uranium is considered to be a nonrenewable energy source, even though it is a common metal found in rocks worldwide. Uranium, referred to as U-235

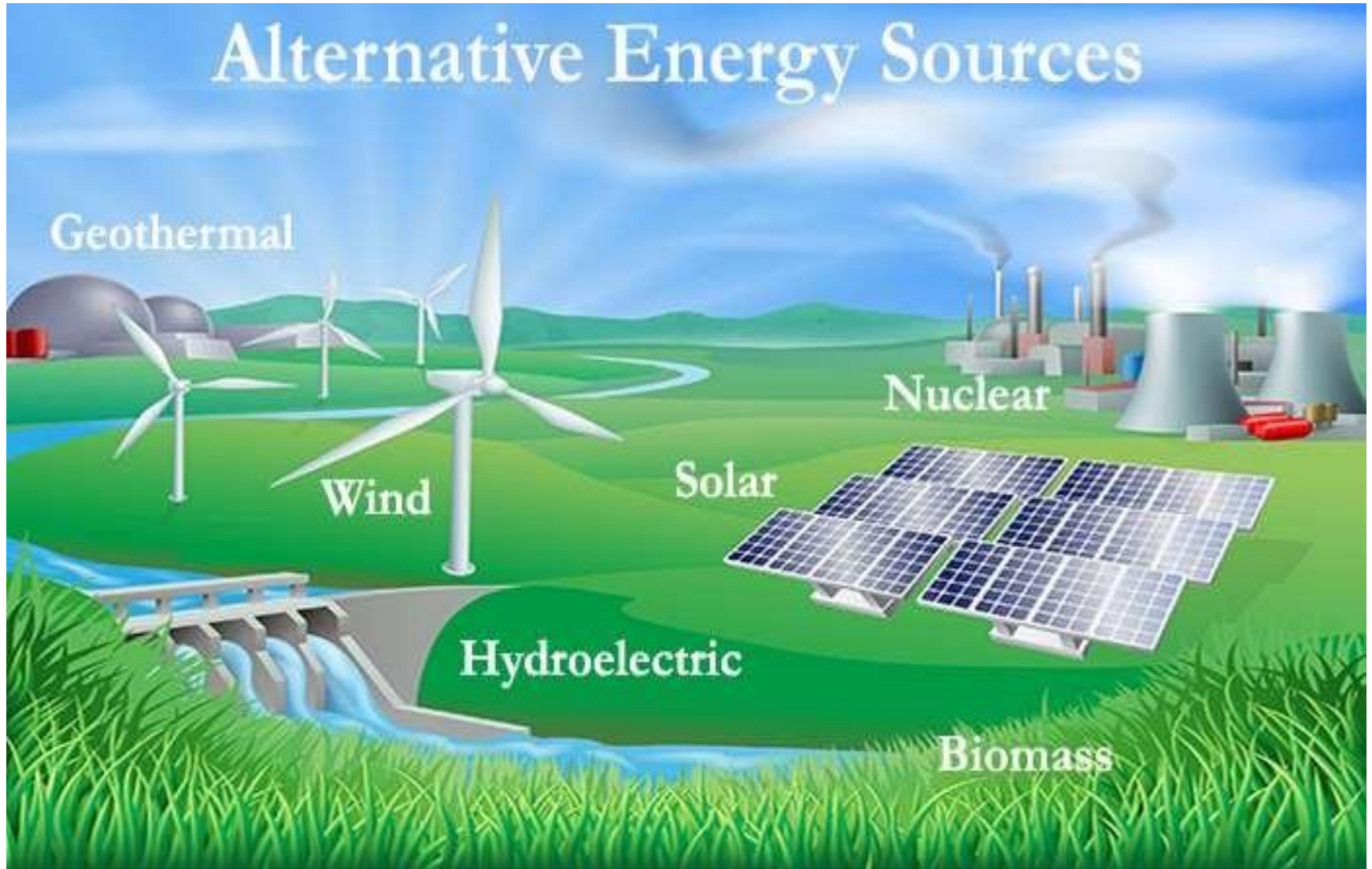
Advantages of Non Renewable

- Non-renewable energy sources are abundant and affordable
- Non-renewable energy is cost effective and easier to use.

Disadvantages of Non Renewable

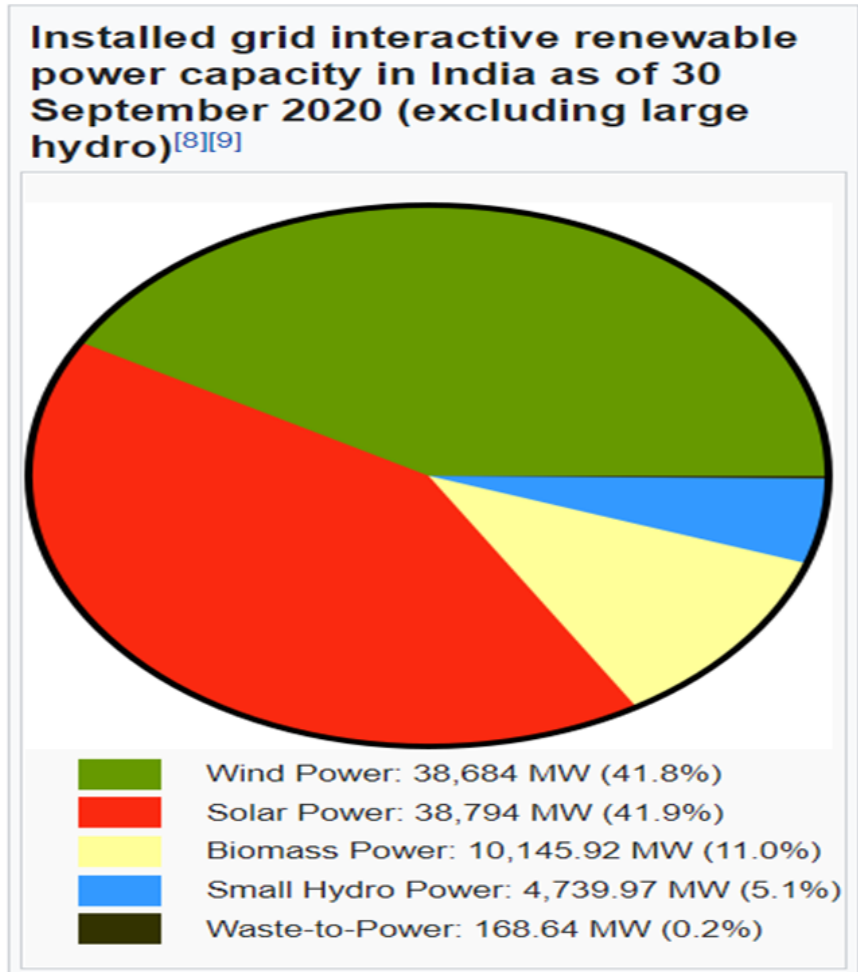
- Once sources of non-renewable energies are gone they can't be replaced or revitalized
- The mining of non-renewable energy and the by-products they leave behind causes damage to the environment
- The burning of fossil fuels continues to rise producing high levels of carbon dioxide (CO₂) which climatologists believe is a major cause of global warming.

Alternate sources of Energy



Total current Renewable energy of India:

- As of 27
- As of November 2020, 38% of India's installed electricity generation capacity is from renewable sources (136 GW out of 373 GW). ..



Total current Renewable energy of India:

Installed grid interactive renewable power capacity (excluding large hydropower) as of 28 February 2021

Source	◆ Total Installed Capacity (MW) ◆	2022 target (MW) ◆
Solar power	38,794	100,000
Wind power	38,684	60,000
Biomass power (Biomass & Gasification and Bagasse Cogeneration)	10,145.92	*10,000
Waste-to-Power	168.64	
Small hydropower	4,739.97	5,000
TOTAL	89,229.42	175,000

Energy Conservation:

- Energy conservation means more efficient utilization of energy resource ensuring the same level of economic activities with lesser inputs of energy.

Measures for Energy Conservation

- 1. Creating Awareness.**
- 2. Improve efficiency of equipment , machinery and appliances.**
- 3. Non- renewable sources should only be used when the substitutes are not available.**
- 4. Alternatives should be made available.**
- 5. Budget allotment should be made for energy conservation plans.**
- 6. Improvise management and maintenance to avoid losses.**
- 7. Need for modernization and renovation of plants.**

Smart-grid Technology

- A smart grid is an electrical grid that uses information and communications technology to gather and act on information, such as information about the behaviors of suppliers and consumers, in an automated fashion to improve the efficiency, reliability, economics, and sustainability of the production and distribution of electricity.

Go Green Save Earth

Save energy.
Save money.
We can help you get started.

